

Project Report

DataCo Supply Chain Analysis

Data Analytics Project

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About Project

This project is a comprehensive supply chain data analysis based on the **DataCo Supply Chain Dataset**, sourced from **Kaggle**. The dataset was selected to enhance and demonstrate practical data analysis skills by working with real-world business variables such as order details, sales, customer demographics, shipping logistics, and profitability.

The objective was to uncover actionable insights that reflect common challenges and opportunities in supply chain management. Through data exploration, cleaning, and visualization, this project aims to identify key trends, performance indicators, and operational bottlenecks that can guide strategic decision-making.

This analysis not only serves as a skills-building exercise but also simulates the role of a data analyst in deriving value from business data, with a focus on improving efficiency, customer satisfaction, and overall profitability.

Data Overview

The dataset used in this project is titled **"DataCo Supply Chain Dataset"**, originally obtained from **Kaggle**, a popular platform for open-source datasets and data science competitions. It is a simulated yet realistic dataset designed to represent the operational data of a large-scale retail supply chain.

The dataset consists of over **1.6 Lakh+ rows** and **53 Columns** and includes detailed information across various dimensions such as:

- **Order Details**: Order ID, order status, dates, and processing time
- **Customer Information**: Names, locations, segment classifications
- **Product Data**: Categories, sub-categories, pricing, and discounts
- **Sales & Profit Metrics**: Revenue, profit, cost, and profit margins
- **Shipping Logistics**: Delivery modes, shipping dates, and delivery duration

The data was downloaded as a structured **CSV file**, which makes it suitable for analysis using tools like Excel, Python (Pandas), or data visualization platforms such as Power BI or Tableau.

Prior to analysis, the dataset underwent a **data preparation phase**, which included handling missing values, converting date formats, filtering irrelevant fields, and ensuring consistency across categorical variables. This helped in maintaining the quality and reliability of the insights derived from the analysis.

This dataset provides a comprehensive foundation to explore various aspects of supply chain performance and supports meaningful business insights through structured analysis.

Objective

The primary objective of this project is to analyze a real-world-inspired supply chain dataset in order to extract **actionable business insights** that can support data-driven decision-making.

Specifically, the project aims to:

- Understand sales performance across product categories, regions, and customer segments
- Identify the most profitable products, customers, and shipping methods
- Detect patterns, trends, and anomalies in order processing and delivery
- Evaluate the impact of discounts, delivery time, and returns on profitability
- Segment customers based on behavior to support targeted strategies
- Present findings through clear visualizations and performance metrics (KPIs)

By fulfilling these objectives, the project demonstrates the practical application of data analysis in solving real business problems and contributes to improving operational efficiency, customer satisfaction, and revenue growth in a supply chain context.

Key Requirments

1. Business Insight Goals

- Identify top-performing products, categories, and regions.
- Analyze customer segments based on purchasing behavior.
- Evaluate shipping efficiency and its impact on delivery timelines and profits.
- Detect trends in returns, cancellations, and seasonal order fluctuations.

2. Data Preparation

- Handle missing or inconsistent data.
- Standardize date formats for accurate time-based analysis.
- Remove irrelevant or duplicate records.
- Convert categorical variables for better grouping and visualization.

3. Analytical Metrics (KPIs)

- Total Sales & Total Profit
- Average Profit Margin
- Sales by Category/Sub-category
- Average Delivery Time
- Return/Cancellation Rate
- Customer Lifetime Value (if possible)

4. Tools & Technologies

- **Data Cleaning & Analysis**: Excel / Power BI
- **Visualization**: Power BI

5. Final Deliverables

- An insight-rich analytical report
- Visual dashboards with interactive elements (if applicable)
- Executive summary with clear business recommendations
- Supporting files (code, Excel sheets, visuals)

Extracted Insights

1. Sales Performance

What are the total sales and profits by product category and sub-category?

sales by category	
Row Labels	**Sum of Sales**
Accessories	56955.82
As Seen on TV!	8222.05
Baby	7633.52
Baseball & Softball	41534.78
Basketball	16431.54
Books	7902.90
Boxing & MMA	36420.10
Cameras	163105.08
Camping & Hiking	2523895.80
Cardio Equipment	1584425.25
CDs	2050.24
Children's Clothing	146898.15
Cleats	1913998.01
Computers	422782.50
Consumer Electronics	65472.97
Crafts	133455.57
DVDs	48203.41
Electronics	161008.59
Fishing	4262946.95
Fitness Accessories	17095.91
Garden	158679.38
Girls' Apparel	63036.80
Golf Apparel	19068.66
Golf Bags & Carts	7221.98
Golf Balls	33101.46
Golf Gloves	58253.74
Golf Shoes	44694.92
Health and Beauty	58433.66
Hockey	23121.14
Hunting & Shooting	24170.19
Indoor/Outdoor Games	1235297.71
Kids' Golf Clubs	55624.05
Lacrosse	16392.64
Men's Clothing	25977.67

Profit by category	
Row Labels	**Sum of Profit Per order**
Accessories	24701.64
As Seen on TV!	3232.67
Baby	2110.00
Baseball & Softball	17232.25
Basketball	5060.80
Books	2254.93
Boxing & MMA	15811.09
Cameras	47304.18
Camping & Hiking	740034.26
Cardio Equipment	666513.29
CDs	588.50
Children's Clothing	43717.63
Cleats	803701.31
Computers	124097.98
Consumer Electronics	19130.32
Crafts	39408.86
DVDs	13829.05
Electronics	68288.57
Fishing	1256368.86
Fitness Accessories	6329.97
Garden	48370.35
Girls' Apparel	27376.63
Golf Apparel	6138.78
Golf Bags & Carts	2121.51
Golf Balls	13423.57
Golf Gloves	21944.21
Golf Shoes	20404.65
Health and Beauty	17135.72
Hockey	8813.04
Hunting & Shooting	9633.77
Indoor/Outdoor Games	521741.49
Kids' Golf Clubs	17490.38
Lacrosse	7269.20
Men's Clothing	7444.00

Men's Footwear	1789207.16
Men's Golf Clubs	26350.11
Music	72545.18
Pet Supplies	24060.76
Shop By Sport	564039.70
Soccer	13263.49
Sporting Goods	71563.84
Strength Training	33917.69
Tennis & Racquet	19242.59
Toys	3956.37
Trade-In	32819.67
Video Games	19364.83
Water Sports	1912985.53
Women's Apparel	1365544.25
Women's Clothing	90565.65
Women's Golf Clubs	20470.80
Grand Total	**19503410.79**

Men's Footwear	531288.62
Men's Golf Clubs	8951.61
Music	21225.35
Pet Supplies	6998.90
Shop By Sport	234425.56
Soccer	4882.76
Sporting Goods	20985.51
Strength Training	9505.34
Tennis & Racquet	7829.19
Toys	1215.43
Trade-In	11894.65
Video Games	5763.68
Water Sports	558792.72
Women's Apparel	573509.85
Women's Clothing	26717.68
Women's Golf Clubs	7550.01
Grand Total	**6660560.32**

Sales
19503410.79

Profit
6660560.32

Which regions/states are contributing most to revenue and profit?

Profit by Region	
Region	**Sum of Profit**
Canada	34594.14
Caribbean	300952.52
Central Africa	60074.88
Central America	1023108.04
Central Asia	19500.96
East Africa	68013.71
East of USA	253349.27
Eastern Asia	265227.95
Eastern Europe	139018.02
North Africa	111631.69
Northern Europe	396739.80
Oceania	368733.70
South America	542001.68
South Asia	280208.99
South of USA	143811.15
Southeast Asia	344677.04

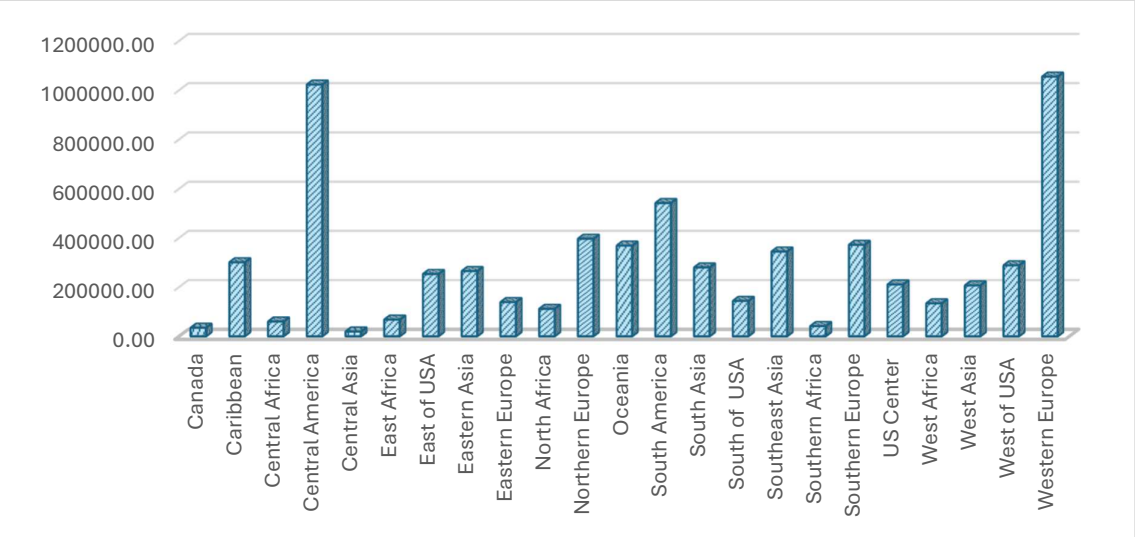
Top 10 Region Profit	
Region	**Sum of Profit**
Western Europe	1055314.48
Central America	1023108.04
South America	542001.68
Northern Europe	396739.80
Southern Europe	371966.12
Oceania	368733.70
Southeast Asia	344677.04
Caribbean	300952.52
West of USA	288571.52
South Asia	280208.99
Grand Total	**4972273.89**

Top 10 States Profit	
States	**Sum of Profit**
Inglaterra	275533.69
Isla de Francia	186747.03

Southern Africa	41332.40
Southern Europe	371966.12
US Center	210674.51
West Africa	133967.89
West Asia	207089.86
West of USA	288571.52
Western Europe	1055314.48
Grand Total	**6660560.32**

California	180797.14
Renania del Norte-Westfalia	123563.74
San Salvador	110115.28
Nueva York	97353.86
Distrito Federal	93464.88
Nueva Gales del Sur	91110.92
Texas	86987.37
Santo Domingo	78669.39
Grand Total	**1324343.30**

Profit by Region



1. Most profitable Region – Western Europe.
2. Most profitable Country – Estados Unidos.
3. Most profitable State – Inglaterra.

2. Customer Segmentation

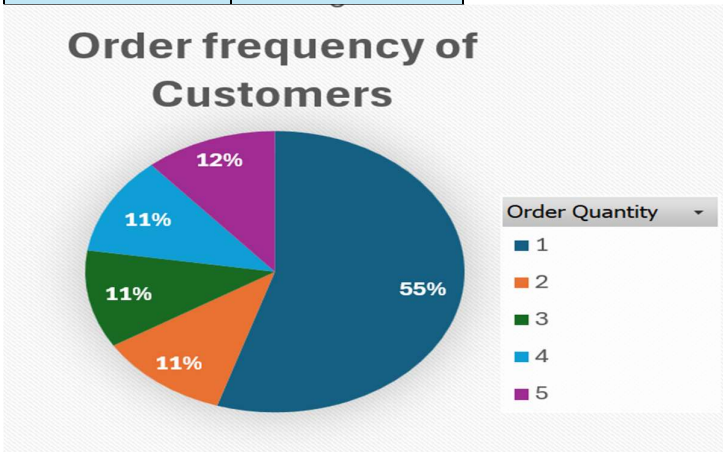
Segment customers based on their order frequency, spending habits, and locations.

Top 10 States by Orders	
States	**Sum of Orders**
Inglaterra	5446
California	3996
Isla de Francia	3715
Renania del Norte-Westfalia	2584
San Salvador	2423
Nueva York	2206
Distrito Federal	2076
Texas	1951
Nueva Gales del Sur	1896
Queensland	1783
Grand Total	**28076**

Top 10 Country by Orders	
Country	**Sum of Orders**
Estados Unidos	20038
México	10739
Francia	10736
Alemania	7572
Australia	6829
Brasil	6466
Reino Unido	5916
China	4653
Italia	4045
India	3828
Grand Total	**80822**

Top 10 most spending Customers	
Customer ID	**Sum of Sales**
2641	5857.18
664	5584.06
5216	5496.76
11431	5297.56
1464	5194.69
749	5190.78
2292	5154.64
10591	5148.40
791	5063.11
3708	5058.28
Grand Total	**53045.46**

Order Frequency and Customers	
Order Frequency	**Customers**
1	79944
2	16399
3	16366
4	16414
5	16428
Grand Total	**145551**



1. Most of the customers orders only one product on a single order.

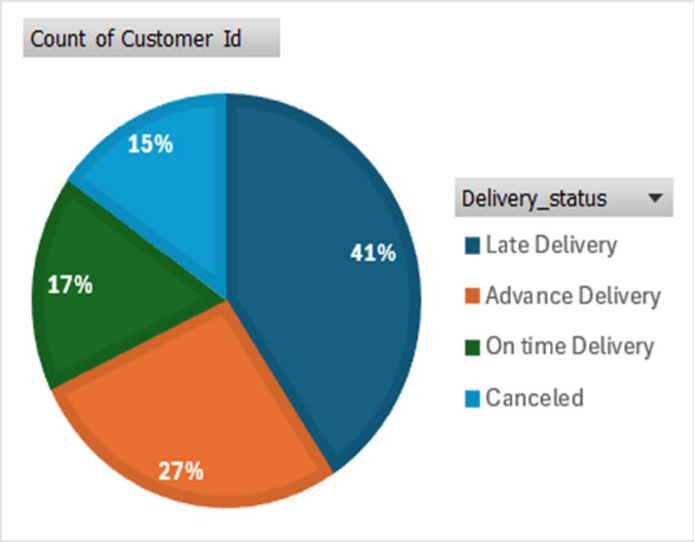
2. Customer ID 2641 is our most valueable customer.

3. Shipping & Delivery Trends

What are the Delivery Status?

Delivery time and Delivery count	
Days of Shippin	Count of Customer Id
1	18782
2	18686
3	18593
4	18684
5	18491
6	10738
7	10948
8	10985
9	10830
10	1461
11	1454
12	1449
13	1422
14	1416
15	312
16	285
17	288
18	284
19	290
20	12
21	19
22	15
23	10
24	12
25	7
26	12
27	11
28	17
29	15
30	7
31	16
Grand Total	**145551**

Delivery Status	
Row Labels	Count of Customer Id
Late Delivery	59227
Advance Delivery	39649
On time Delivery	24741
Canceled	21934
Grand Total	**145551**



1. 44.23% Orders Delivers on time
2. 41% Orders Delivers late
3. 15% Orders got canceled

4. Profitability Analysis

Which products or categories are yielding the highest and lowest profit margins?

Top 5 most Demanded Category	
Product Category	**Sum of Profit**
Fishing	1256368.86
Cleats	803701.31
Camping & Hiking	740034.26
Cardio Equipment	666513.29
Women's Apparel	573509.85
Grand Total	**4040127.57**

Top 5 least Demanded Category	
Product Category	**Sum of Profit**
CDs	588.50
Toys	1215.43
Baby	2110.00
Golf Bags & Carts	2121.51
Books	2254.93
Grand Total	**8290.37**

Top 10 most profitable Products	
Product Name	**Sum of Profit**
Field & Stream Sportsman 16 Gun Fire Safe	1256368.86
Perfect Fitness Perfect Rip Deck	801970.25
Diamondback Women's Serene Classic Comfort Bi	740034.26
Nike Men's Free 5.0+ Running Shoe	661260.65
Nike Men's Dri-FIT Victory Golf Polo	573509.85
Pelican Sunstream 100 Kayak	556465.64
Nike Men's CJ Elite 2 TD Football Cleat	531288.62
O'Brien Men's Neoprene Life Vest	521741.49
Under Armour Girls' Toddler Spine Surge Runni	227747.75
Dell Laptop	124097.98
Grand Total	**5994485.35**

Top 5 most Demanded product Category and products name	
Product Category	**Sum of Profit**
Fishing	**1256368.86**
Field & Stream Sportsman 16 Gun Fire Safe	1256368.86
Cleats	**803701.31**
Perfect Fitness Perfect Rip Deck	801970.25
Total Gym 1400	1731.06
Camping & Hiking	**740034.26**
Diamondback Women's Serene Classic Comfort Bi	740034.26
Cardio Equipment	**666513.29**
Nike Men's Free 5.0+ Running Shoe	661260.65
Nike Women's Tempo Shorts	5252.64
Women's Apparel	**573509.85**
Nike Men's Dri-FIT Victory Golf Polo	573509.85
Grand Total	**4040127.57**

Correlation between Discount and Profit

Count of orders by Discount	
Discount %	**Count of Customer Id**
0%	8108
1%	8114
2%	8128
3%	8024
4%	8065
5%	8083
6%	8148
7%	8071
9%	8157
10%	8127
12%	8065
13%	8050
15%	8085
16%	8074
17%	8015
18%	8123
20%	8000
25%	8114
Grand Total	**145551**

Profit by Discount	
Discount %	**Sum of Profit Per order**
0%	416116
1%	408089
2%	401498
3%	390444
4%	392675
5%	391325
6%	388273
7%	384315
9%	385326
10%	375039
12%	359572
13%	357368
15%	342426
16%	346860
17%	340703
18%	338978
20%	327925
25%	313628
Grand Total	**6660560**

1. Most demanded category – Fishing.
2. Least demanded category – CDs.
3. Most demanded product - Field & Stream Sportsman 16 Gun Fire Safe
4. There are no relation between Discount and Profit.

5. Order Trends

Identify monthly/quarterly trends in orders.

Profit by Year and most Demanded category	
Year with Demanded product	**Sum of Profit Per order**
2014	**13437.52**
Fishing	5333.42
Cleats	4359.32
Camping & Hiking	3744.78
2015	**958368.74**
Fishing	426057.20
Cleats	278173.13
Camping & Hiking	254138.41
2016	**977232.42**
Fishing	440290.09
Cleats	279787.64
Camping & Hiking	257154.69
2017	**733022.60**
Fishing	330878.15
Cleats	207723.77
Camping & Hiking	194420.68
2018	**33411.41**
Women's Clothing	12908.15
Sporting Goods	10952.49
Garden	9550.77
#VALUE!	**118612.13**
Fishing	53810.00
Cleats	33657.45
Cardio Equipment	31144.68
Grand Total	**2834084.82**

Orders by Month	
Order month	**Total Orders**
January	12972
August	12131
May	12089
March	12080
April	12049
July	11933
September	11833
June	11524
February	11189
October	10834
November	10401
December	10312
#VALUE!	6204
Grand Total	**145551**

Profit by Year	
Order year	**Sum of Profit**
2016	2129549.82
2015	2120645.34
2017	2030009.38
#VALUE!	289272.42
2018	63776.40
2014	27306.96
Grand Total	**6660560.32**

Profit by Year	
Product Category	**Sum of Profit Per order**
2016	**2129549.82**
January	174901.27
February	168014.45
March	177075.30
April	177636.57
May	171695.36
June	178808.84
July	183617.51

August	181004.07
September	181719.83
October	189830.74
November	175395.91
December	169849.97
2015	**2120645.34**
January	180373.22
February	158788.44
March	178505.40
April	178542.38
May	183971.99
June	174159.89
July	175884.92
August	177004.04
September	174166.60
October	180256.66
November	177170.48
December	181821.32
2017	**2030009.38**
January	181106.65
February	185162.69
March	185828.55
April	185272.88
May	176469.90
June	171536.79
July	174013.95
August	179571.21
September	176962.49
October	176115.72
November	131944.34
December	106024.21
#VALUE!	**289272.42**
#VALUE!	289272.42
2018	**63776.40**
January	26618.00
February	6402.19
March	5251.72
April	5070.18
May	5402.87
June	4066.53
July	3811.78
August	1666.83
September	1361.43
October	1827.34
November	2297.53
2014	**27306.96**
December	27306.96
Grand Total	**6660560.32**

1. Most profitable Year - 2016

2. Most profitable Month - Jan

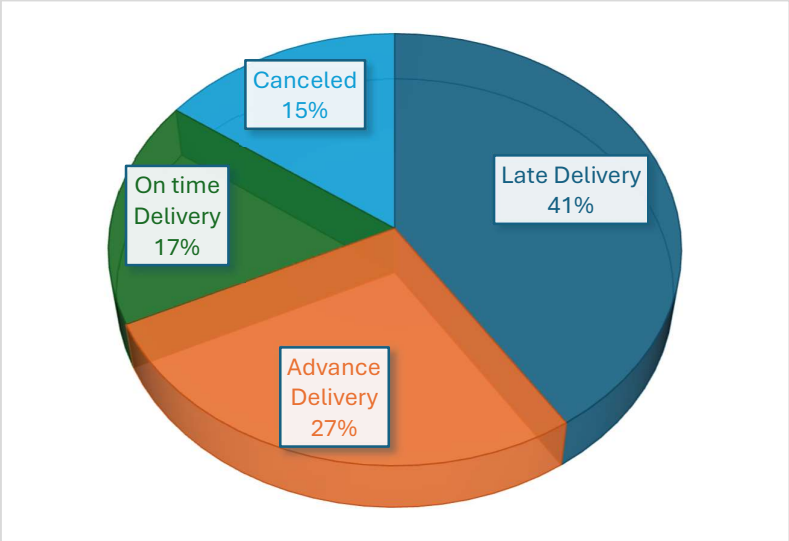
3. here the faulted value follows the trend of 2014 to 2017 so there are high probability that these data belonges to that trend.

4. Least profitable Year - 2018

6. Returns and Cancelation

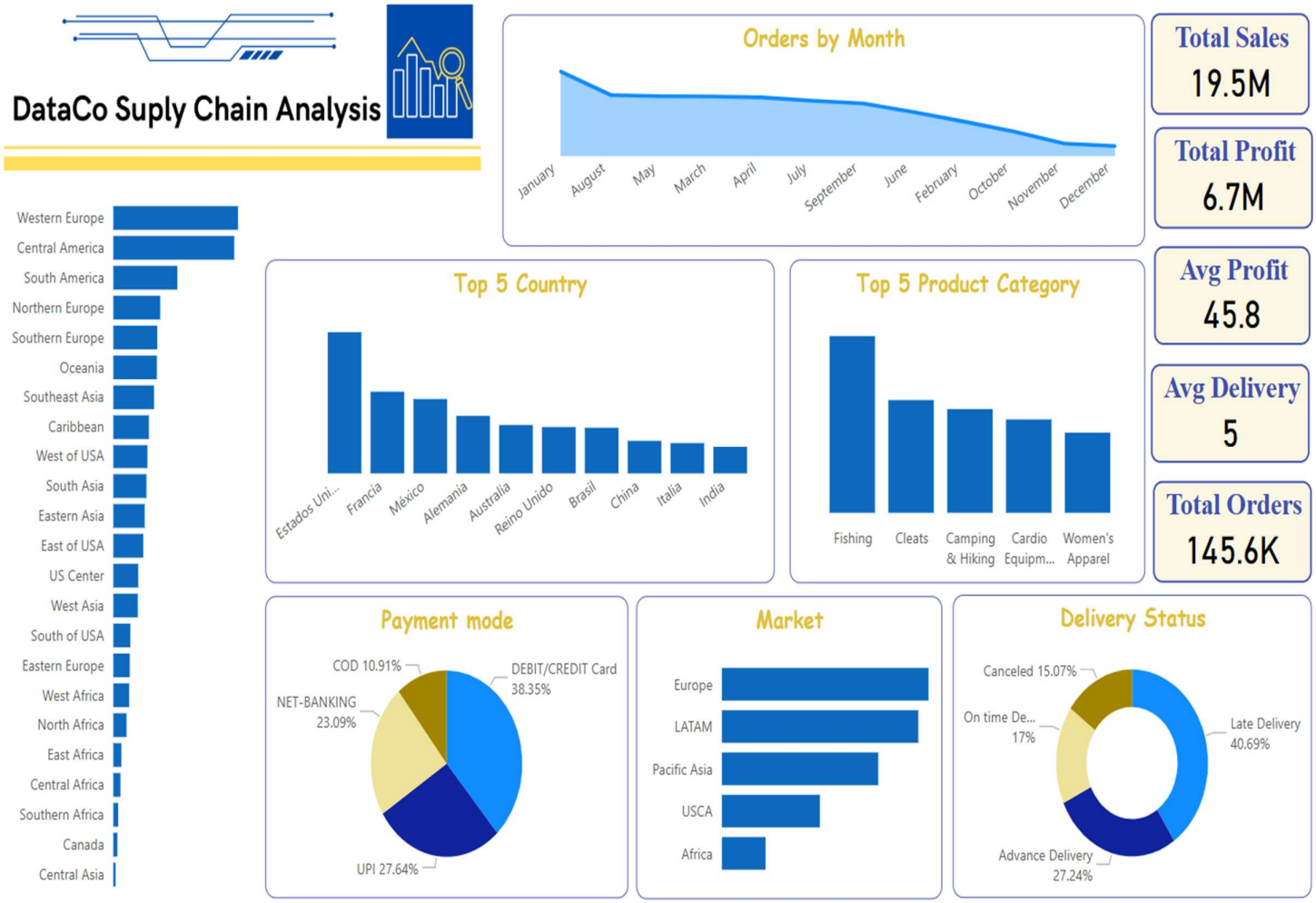
What percentage of orders are returned or canceled?

Delivery status	
Delivery status	**Orders**
Late Delivery	59227
Advance Delivery	39649
On time Delivery	24741
Canceled	21934
Grand Total	**145551**



1. 44.23% Orders Delivers on time
2. 41% Orders Delivers late
3. 15% Orders got canceled

Dashboard


DataCo Suply Chain Analysis
Orders by Month
January
August
May
March
April
July
September
June
February
October
November
December
Total Sales
19.5M
Total Profit
6.7M
Avg Profit
45.8
Avg Delivery
5
Total Orders
145.6K
Western Europe
Central America
South America
Northern Europe
Southern Europe
Oceania
Southeast Asia
Caribbean
West of USA
South Asia
Eastern Asia
East of USA
US Center
West Asia
South of USA
Eastern Europe
West Africa
North Africa
East Africa
Central Africa
Southern Africa
Canada
Central Asia
Top 5 Country
Estados Uni...
Francia
México
Alemania
Australia
Reino Unido
Brasil
China
Italia
India
Top 5 Product Category
Fishing
Cleats
Camping & Hiking
Cardio Equipm...
Women's Apparel
Payment mode
COD 10.91%
DEBIT/CREDIT Card 38.35%
NET-BANKING 23.09%
UPI 27.64%
Market
Europe
LATAM
Pacific Asia
USCA
Africa
Delivery Status
Canceled 15.07%
On time De... 17%
Late Delivery 40.69%
Advance Delivery 27.24%

Data Gathering

